CISC 5640 Redis Assignment

Part 1: Setup

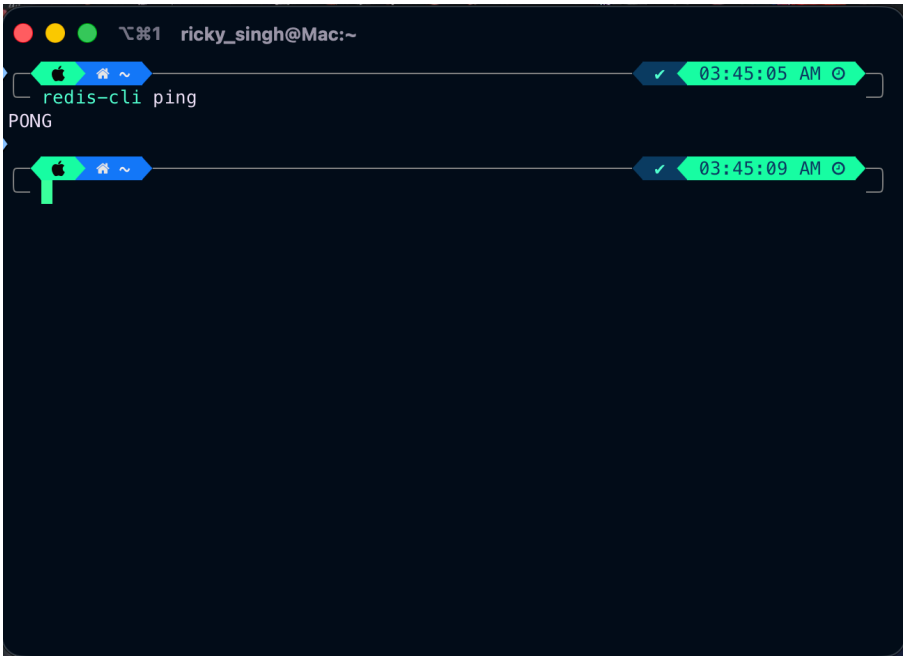


Figure 1: Terminal output verifies the successful installation of Redis

Part 2: Redis Basics



Figure 2: Terminal output shows a user profile has being set as a single value



Figure 3: Terminal output shows a user profile hash fields being set in separate instances

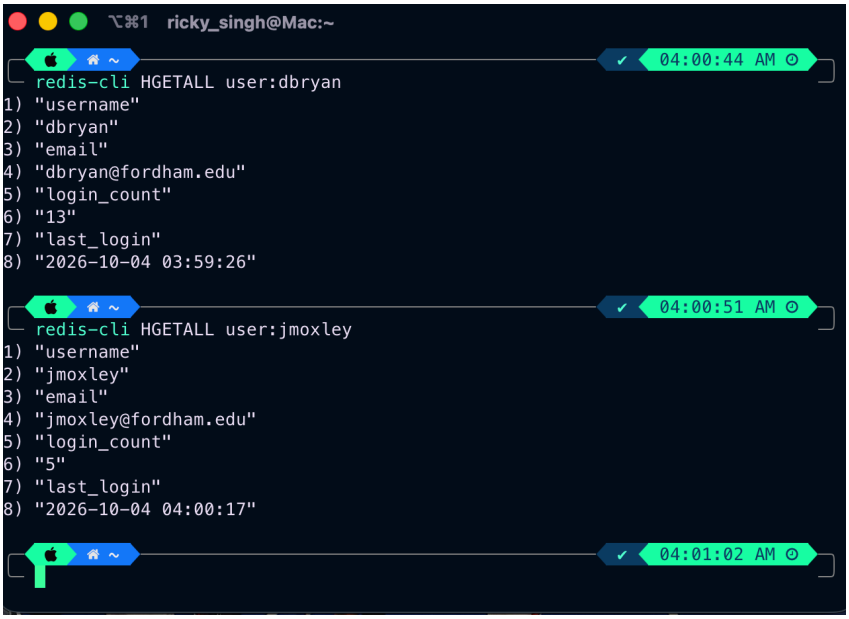


Figure 4: Terminal shows all the fields for user profiles being outputted using the HGETALL command

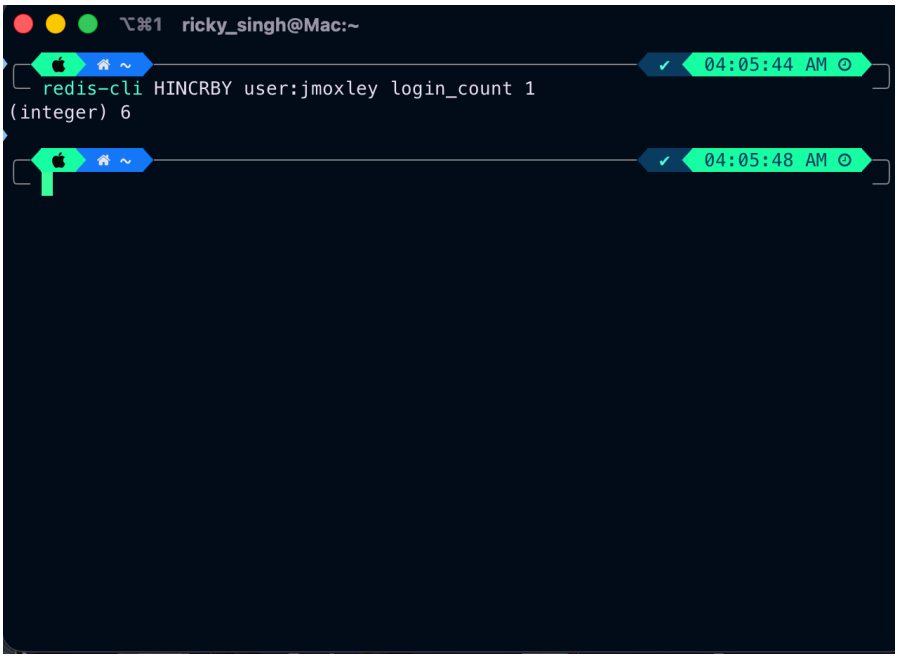


Figure 5: Terminal output shows login_count be incremented by using the HINCRBY command

Part 2: Data Structures

Sets



Figure 6

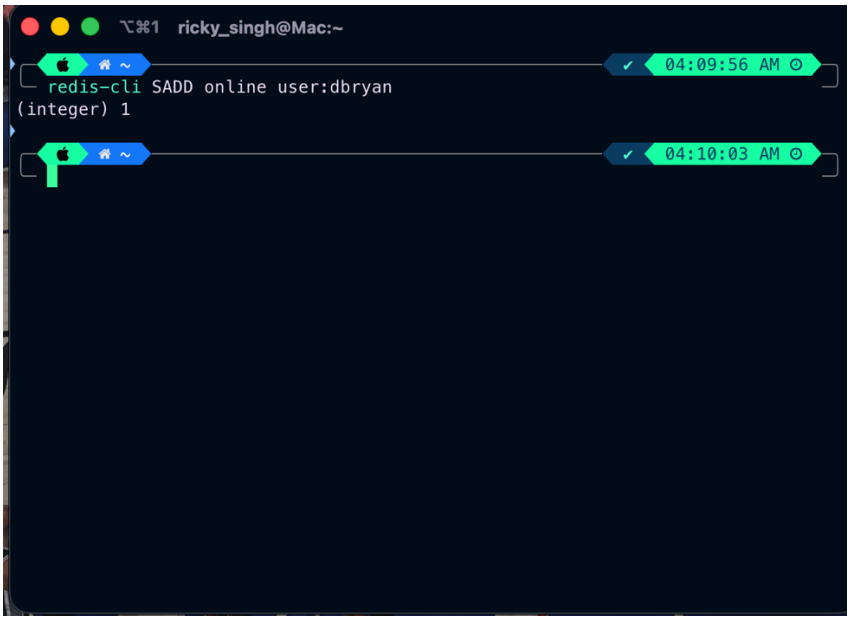


Figure 7: Terminal output shows a member being added to the online set using the SADD command



Figure 8: Terminal output shows the entire set being displayed using SEMEMBERS command, a member being removed using the SREM command, and checking the set after the removal

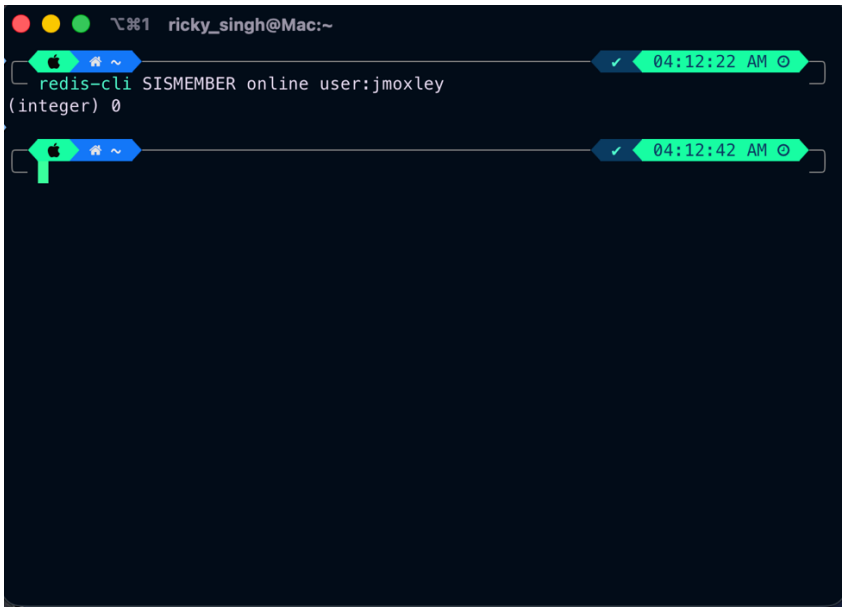


Figure 9: Terminal output shows a profile being checked whether it is a member of the set using the SISMEMBER command

Sorted Sets

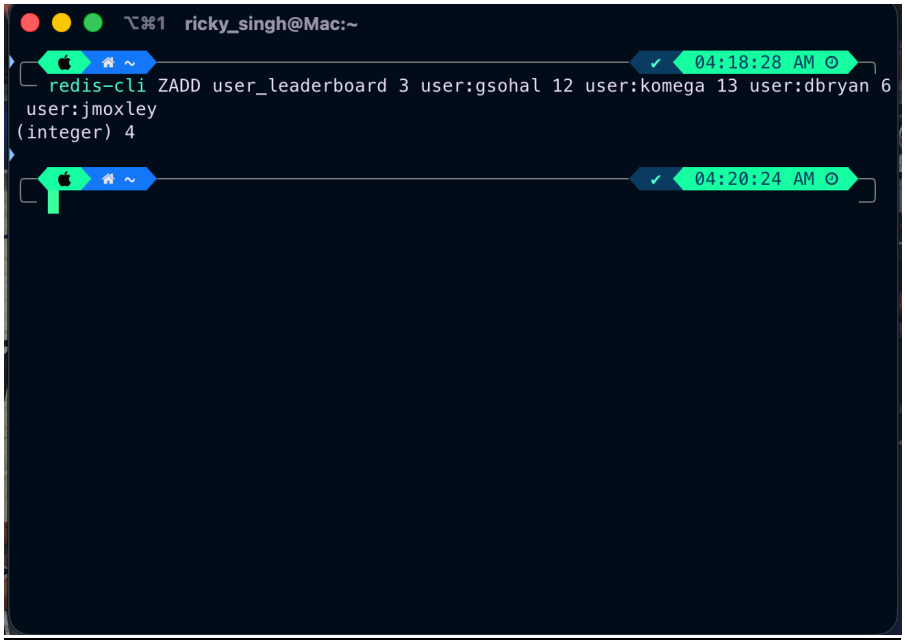


Figure 10: Terminal output shows a sorted set being created using the ZADD command

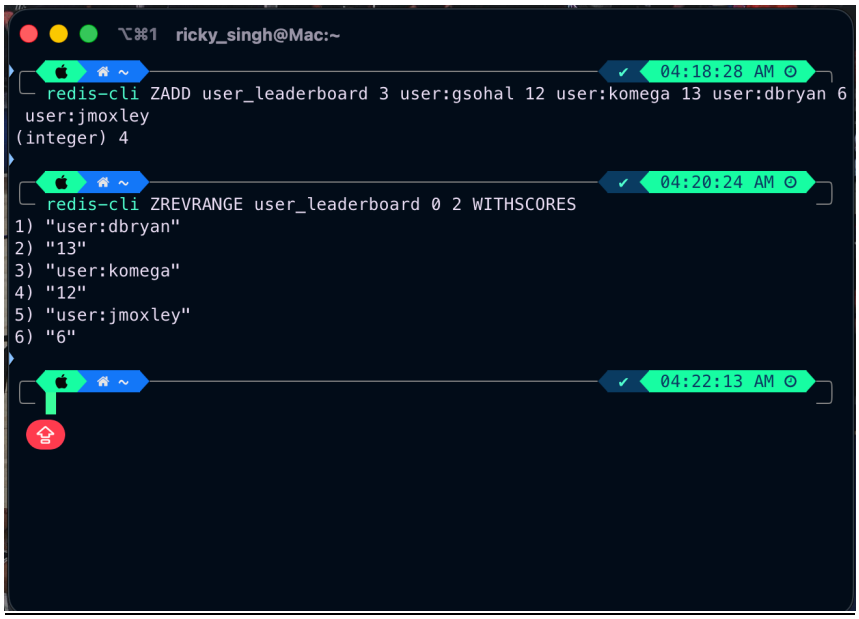


Figure 11: Terminal output shows the user leaderboard showing the top 3 users with their login count

Lists



Figure 12: Terminal output shows multiple values being pushed to the user:komega:actions list using the LPUSH command

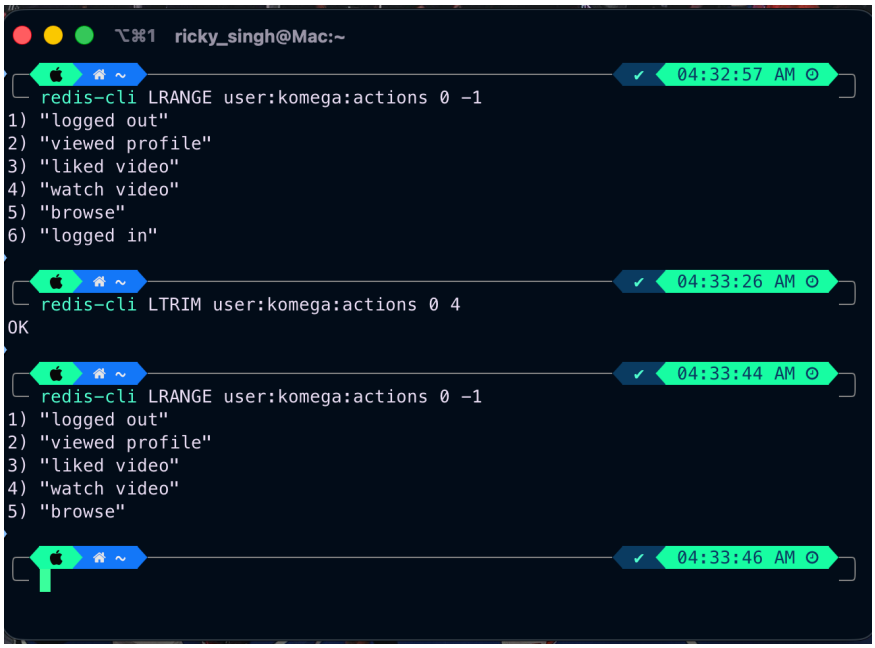


Figure 13: Terminal output shows the user:komega:actions list being trimmed to 5 elements using the LTRIM command

Part 4: Expiration & Persistence

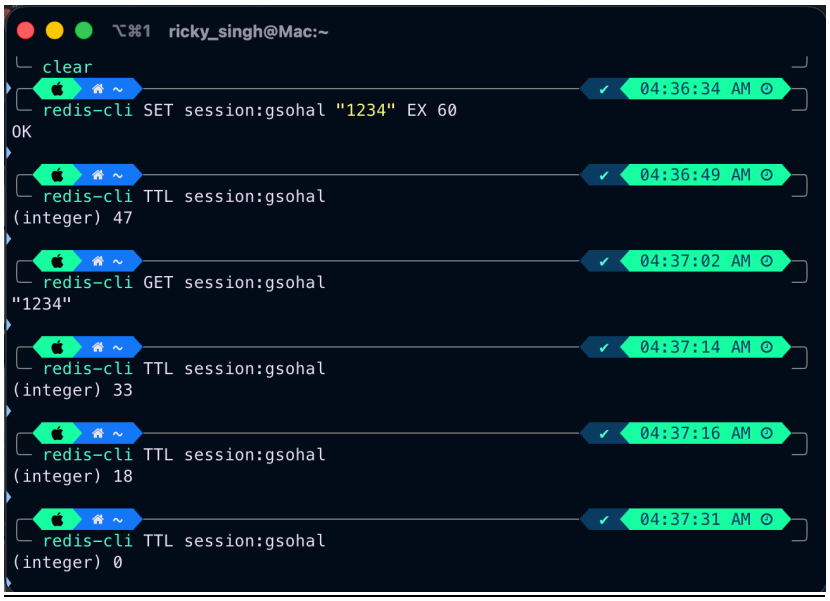


Figure 14: Terminal output shows a session token being set with 60 seconds and the time is checked using the TTL command till it expires

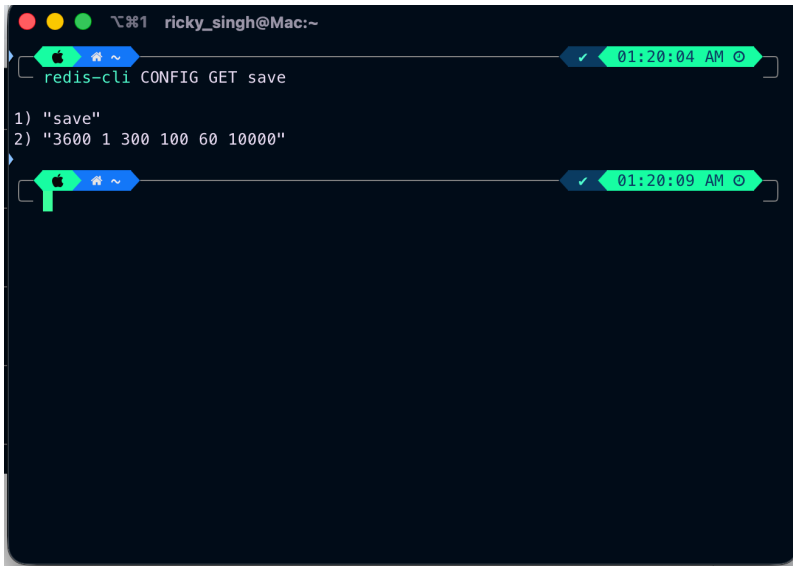


Figure 15: Terminal output verifies snapshot is enabled

Part 5 Reflection & Comparison

Redis is often used as a cache because it is an in-memory datastore. Features that make it well-suited are the fast read and write speeds and the support for data expiration. Redis allows data that is frequently used to be quickly accessed without having to query the database. In addition, data expiration is useful in that cached data may become outdated and does not need to be stored permanently. A user can set a cached data to expire after X seconds so that after the application can retrieve new data from the database.

The limitations of Redis compared with MongoDB or Cassandra are its dependency on RAM. With Redis, one is expected to cache a small amount of data. However, caching a large amount of data is not advised as it would become quite expensive. In addition, data can be lost if it is not properly configured for persistence. MongoDB is a document-oriented database that is well suited to store complex data as opposed to Redis. In addition, Cassandra is designed for handling large amounts of data that tends to be scaled across many servers while Redis is suited for situations where fast access to memory is more important than storing large amounts of data. An example where Redis could play a useful role is a medical application, in which the patients lab results are quickly accessed with querying the database